Information Architecture

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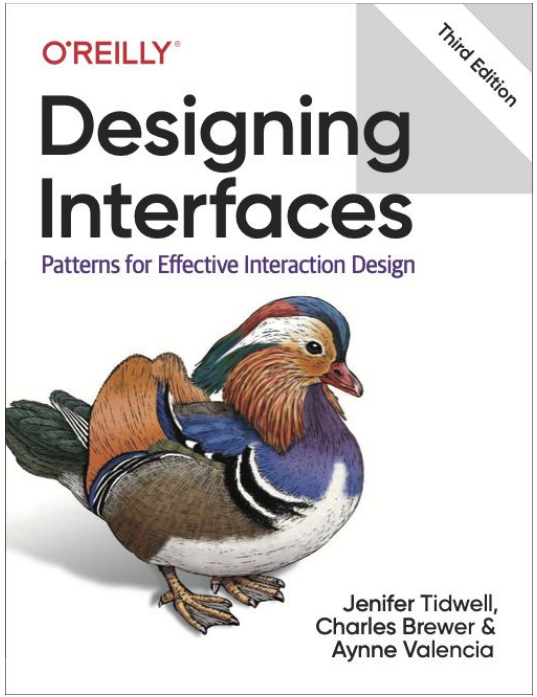
References



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GUI Design

Graphical User Interface (GUI) Design focuses on anticipating what users might need to do and ensuring that the interface has elements that are easy to access, understand, and use to facilitate those actions.

UI brings together concepts from **interaction design, visual design, and information architecture.**

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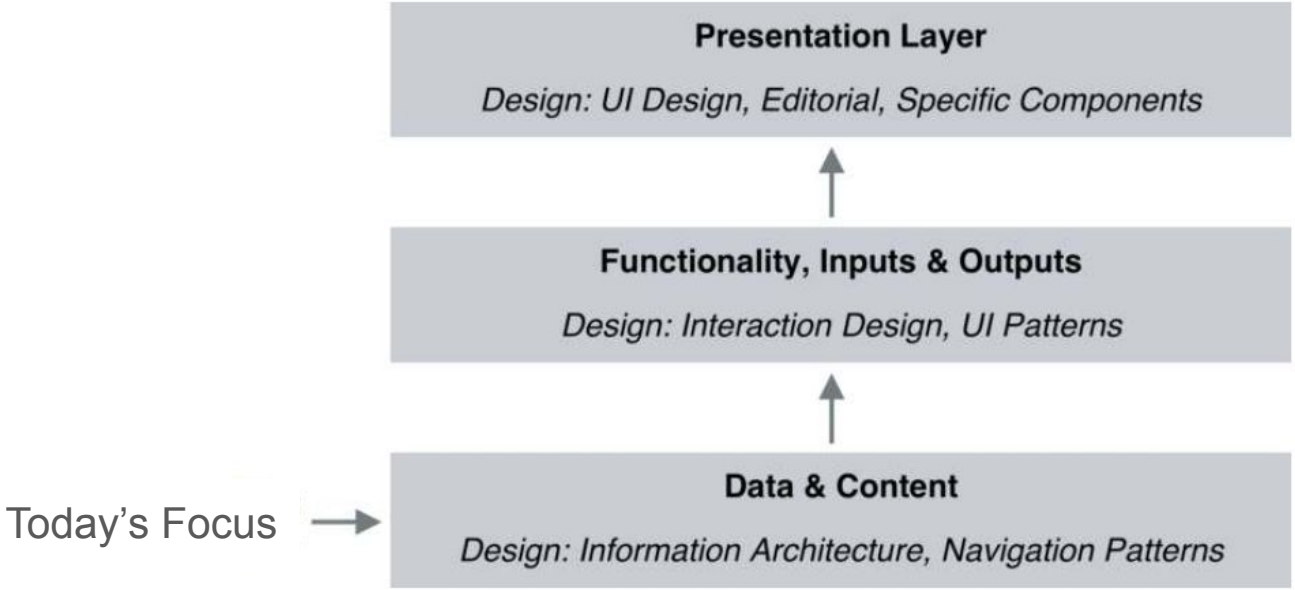
Layered Design



To design this layer well, we first need to understand how people behave...

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Users behavior patterns

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Users behavior patterns

We've explored how to understand users through personas and scenarios, identifying their needs, goals, and motivations.

However, when users interact with interfaces, their behavior follows certain predictable patterns. These patterns emerge regardless of the specific interface or user demographics.

Understanding these patterns helps us:

- Design interfaces that feel natural to users
- Support common user strategies and habits
- Avoid fighting against natural user behaviors
- Create more intuitive interactions

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Self Exploration

Let me explore without getting lost or getting into trouble.

- Likely to learn more-and feel more positive about it
- Clicking pop-up windows, re-entering data mistakenly erased, pausing videos
- Make avenues of exploration available without costing the user anything

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Instant Gratification

I want to accomplish something now, not later.

- People like to see immediate results from the actions they take-it’s human nature
- Successful Experience = Gratification
- You should design the UI to make that first thing stunningly easy
- The user feels more confident in themselves

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Instant Gratification

I want to accomplish something now, not later.

- The user feels more confident in the application
- More likely to use it, even if it gets harder later
- Introductory functionality should not be hidden
- e.g. long sets of instructions, slow-to-load screens, or advertisements

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Satisficing

This is good enough. I don’t want to spend more time learning to do it better.

- Pick first thing-even if it might be wrong
- Satisficing = Satisfying + Sufficing
- People Accept “good enough” instead of “best” if alternative costs time and effort
- Make labels short, plainly worded, and quick to read
- Write them so that users first guess about the meaning is correct

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Satisficing

This is good enough. I don’t want to spend more time learning to do it better.

- Provide escape hatches
- Complicated interface imposes a large cognitive cost on new users
- Visual complexity will often tempt non-experts to satisfice
- Satisficing is why many users end up with odd habits after they’ve been using a system for a while

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Changes in Midstream

I changed my mind about what I was doing.

- Provide opportunities for people to change their goals
- Don’t lock users into a choice-poor environment with no global navigation
- Unless there is good reason: Wizard, Hub and Spoke, and Modal Panel
- Start a process, stop in the middle, and come back to it later to pick up where they left off-a property called “reenterance”

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Deferred Choices

I don’t want to answer that now, just let me finish.

- Users often skip seemingly unnecessary questions and come back to it later
- Answer bare min and come back later
- Most users just don’t want to answer the questions at that time
- Others may not have enough information to answer yet
- Present the short list, hide the long list

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Deferred Choices

I don’t want to answer that now, just let me finish.

- Use good defaults, allow users to view defaults
- “You can always change this later by …”
- Some sites store a users half-finished form entries or other persistent data
- Far more likely to register if they’re allowed to experience the website-drawn in and engaged-and then asked later about who they are

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Incremental Construction

That doesn’t look right; let me change it again. That’s better.

- People don’t usually create all at once
- Users start with small pieces, work on it, step back and look at it, test it, fit it and start to build other parts of it
- Builder-style interfaces need to support that style of work

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Incremental Construction

That doesn’t look right; let me change it again. That’s better.

- Keep interface responsive to quick changes and saves
- Feedback is critical: constantly show the user what the whole thing looks and behaves like while the user works
- When good tools support creative activities, the activities can induce a state of “flow” in the user
- The enjoyment of the activity is it’s own reward

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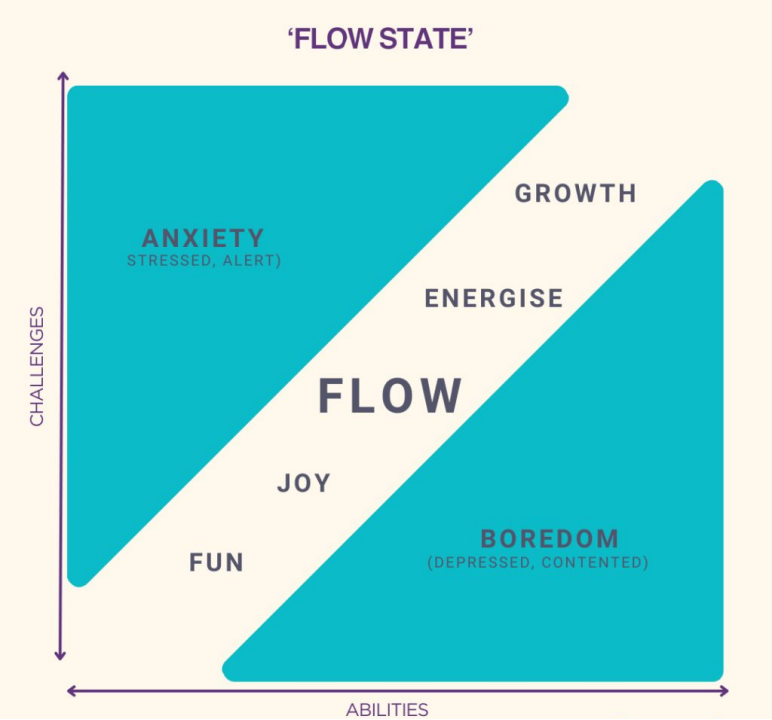
Incremental Construction

That doesn’t look right; let me change it again. That’s better.



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Habituation

That gesture works everywhere else; why doesn’t it work here, too?

- The user no longer has to think about habitual actions
- Habituation measurably improves efficiency, as you can imagine
- But it can also lay traps for the user
- If a gesture doesn’t work-or, worse, does something destructive

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Microbreaks

I’m waiting for the train. Let me do something useful for two minutes.

- Users naturally take small breaks between focused activities
- These short pauses are essential for cognitive processing and stress relief
- Interface design needs to support graceful interruption and resumption
- If tasks can't be easily paused and resumed, users get frustrated and make mistakes

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Spatial Memory

I swear that button was here a minute ago. Where did it go?

- People often find objects by remembering where they are, not what they're named
- Dialog buttons such as "OK" and "Cancel" go in predictable places
- Explains why it's good to provide user-arranged areas for storing objects
- The tops and bottoms of lists and menus are special locations, cognitively speaking

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Prospective Memory

I’m putting this here to remind myself to deal with it later.

- If you need to bring a book to work for the next day, you might place it by the door
- Need to respond to an email, you might leave it on your screen as a reminder
- Give People the tools to create their own reminder systems

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Prospective Memory

I’m putting this here to remind myself to deal with it later.

- If someone leaves a form half-finished, and closes it, you could retain the data
- Think about how to leave some artifacts around, that identify the unfinished tasks

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Streamlined Repetition

I have to repeat this how many times?

- Users sometimes find themselves having to perform the same operation over and over again
- The easier it is for them, the better
- Streamline repetitive tasks that could other-wise be time consuming

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Keyboard Only

Please don’t make me use the mouse.

- Relates to habituation and flow
- Makes it easier to streamline repetitive tasks
- Keyboard commands will always be faster than mouse movements
- Shortcuts are assumed; “control-s”, “tab”, “arrow keys” “return key” “space bar”

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Other People’s Advice

What did everyone else say about this?

- The advice of peers, whether direct or indirect, influences people’s choices
- People can be more effective when aided by others. If not, they might at least be happier with the outcomes
- An online support community is a valuable part of a complete help system

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What We Learned

- Users explore safely when they can't get into trouble
- They want instant results, not long setup
- They satisfice - picking "good enough" over perfect
- They change their minds mid-stream
- They defer decisions they're not ready to make
- They build things incrementally with constant feedback
- They rely on spatial memory and habituation
- They need support for interruptions and resumption

What This Means for Design

Now we need systematic ways to support these behaviors. Information Architecture provides the framework:

- **Organizing** content so users can satisfice with confidence
- **Structuring** navigation to support spatial memory and provide escape routes
- **Labeling** clearly so exploration feels safe
- **Creating** flexible structures where users can change direction

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Information Architecture

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Information Architecture

Information architecture (IA) focuses on organizing, structuring, and labeling content in an effective and sustainable way. The goal is to **help users find information** and complete tasks.

To do this, you need to understand how the pieces fit together to create the larger picture, how items relate to each other within the system.

Why a Well Thought Out IA Matters?

According to Peter Morville Site exit disclaimer, the **purpose of your IA is to help users understand where they are, what they've found, what's around, and what to expect.** As a result, your **IA informs the content strategy** through identifying word choice as well as informing user interface design and interaction design through playing a role in the wireframing and prototyping processes.

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Information Architecture

To be successful, you need a diverse understanding of industry standards for creating, storing, accessing and presenting information. Lou Rosenfeld and Peter Morville in their book, Information Architecture for the World Wide Web, note that the main components of IA are:

- **Organization Schemes and Structures:** How you categorize and structure information
- **Labeling Systems:** How you represent information
- **Navigation Systems:** How users browse or move through information
- **Search Systems:** How users look for information

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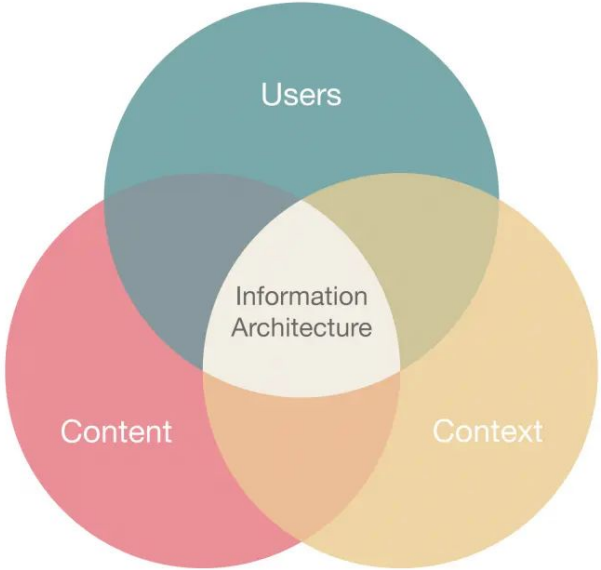
Information Architecture

In order to create these systems of information, you need to understand the interdependent nature of users, content, and context.

Rosenfeld and Morville referred to this as the “information ecology” and visualized it as a venn diagram.

Each circle refers to:

- **Context:** business goals, funding, politics, culture, technology, resources, constraints
- **Content:** content objectives, document and data types, volume, existing structure, governance and ownership
- **Users:** audience, tasks, needs, information-seeking experience



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Organization Schemes

Organization schemes have to do with how you are going to **categorize** your **content** and the various ways you'll create **relationships** between each piece. Most content can be categorized in multiple ways.

Schemes can be broken down into **Exact and Subjective.**

Depending on the content, it’s conceivable that the site may combine schemes as opposed to treating them independently.

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Exact Organization Schemes

Exact organization schemes objectively **divide information into mutually exclusive sections**.

These systems comparatively are easy for information architects to create and categorize content within. However, they can be a challenge at times for users.

It requires that the user understands how what they are looking for fits within the model.

Examples of exact organizational structures include:

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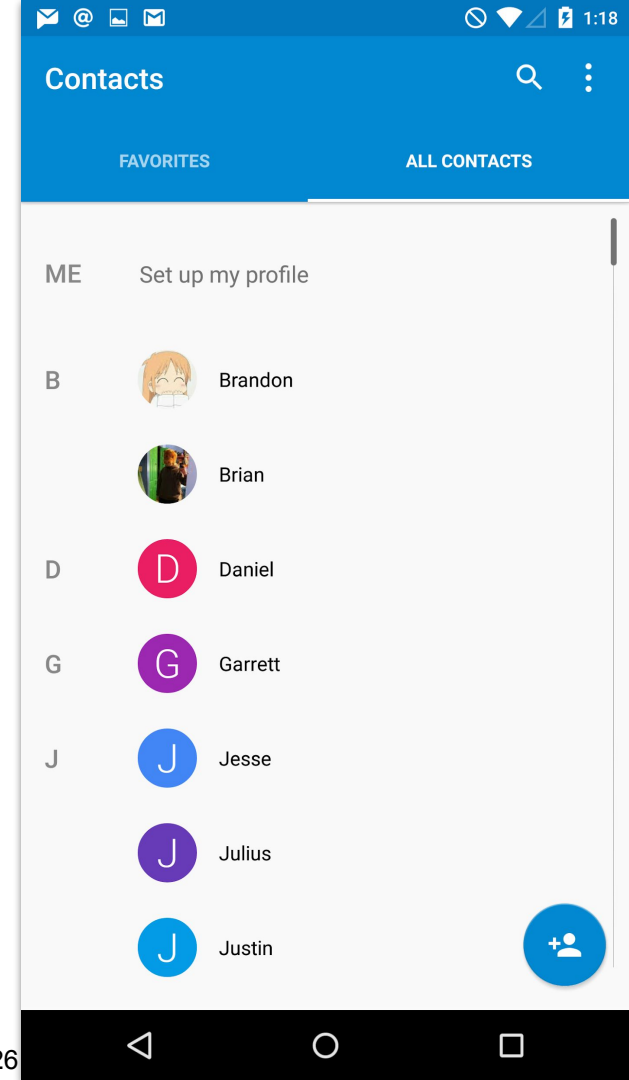
Exact Organization Schemes

Alphabetical schemes make use of our 26-letter alphabet for organizing their contents.

For this type of scheme to be successful, it is important that the content labeling matches the words that users are looking for.

Sometimes, alphabetical schemes in the form of an A-Z index serve as secondary navigational components to supplement content findability that is otherwise organized.

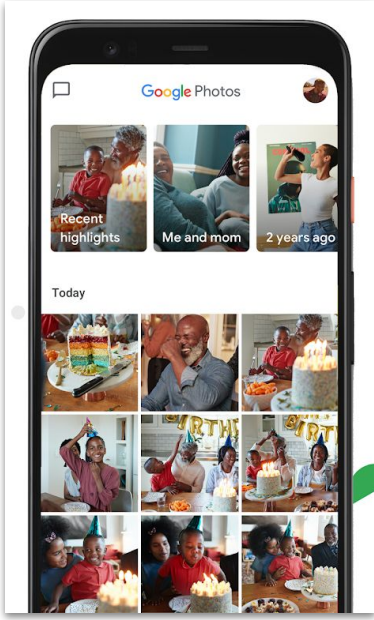


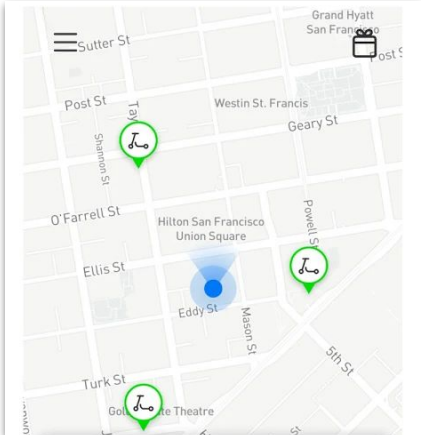
Exact Organization Schemes

Chronological schemes organize content by date. For these schemes to be successful there must be agreement about when the subject of the content took place.

Geographical schemes organize content based on place. Unless there are border disputes, this type of scheme is fairly straightforward to design and use.

Often these types of schemes serve as a good supplemental way to navigate a site that that is otherwise organized. For example, you may choose to provide a map to display information or an A-Z index to get to topics grouped primarily by one of the following subjective schemes.





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Subjective Organization Schemes

Subjective organization schemes **categorize information in a way that may be specific** to or defined by the organization or field.

Although they are difficult to design, **they are often more useful than exact organization schemes.**

When information architects take the time to consider the user's mental models and group the content in meaningful ways, these types of schemes can be quite effective in producing conversions.

This type of categorization can also help facilitate learning by helping users understand and draw connections between pieces of content.

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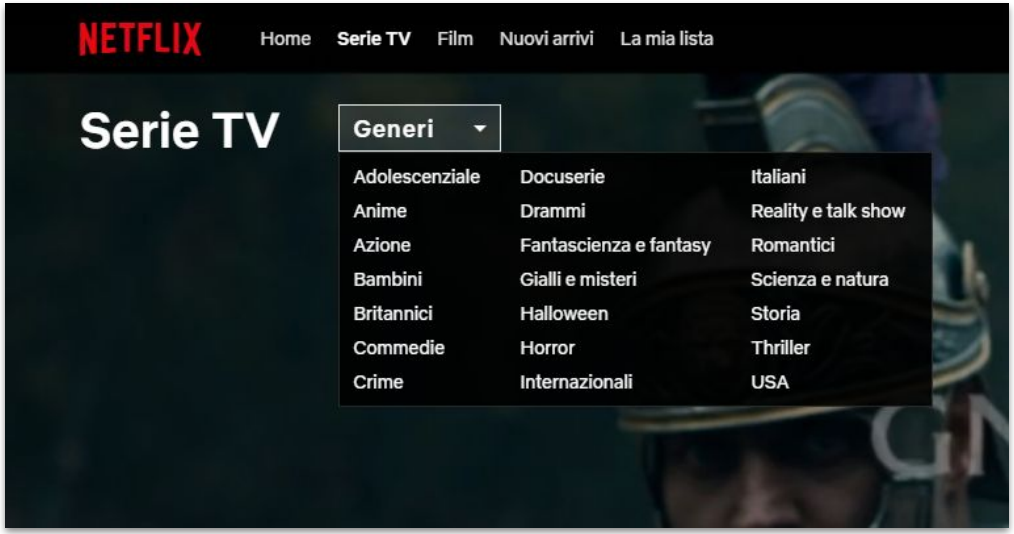
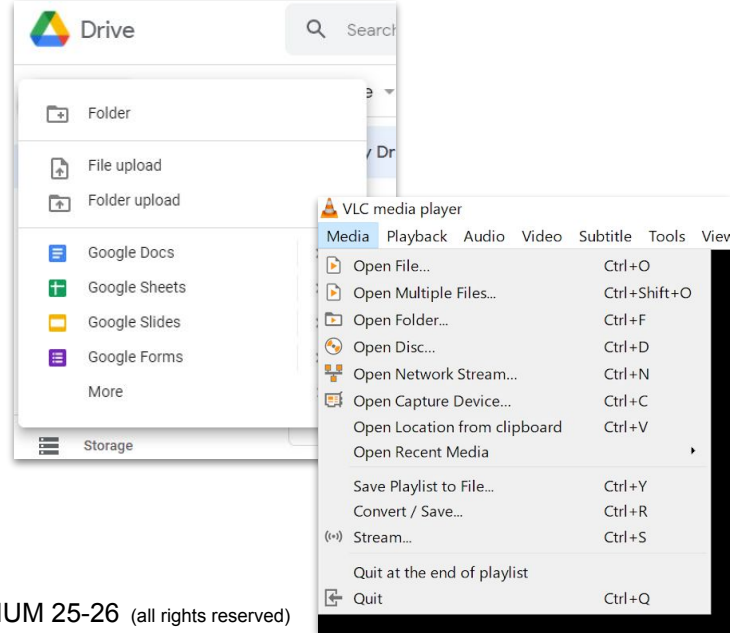
Subjective Organization Schemes

Examples of subjective schemes include the following:

Topic schemes organize content based on the specific subject matter.

Task schemes organize content by considering the needs, actions, questions, or processes that users bring to that specific content.





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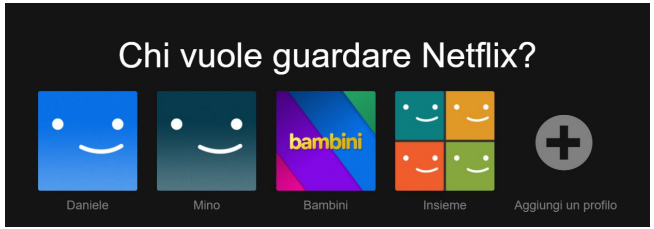
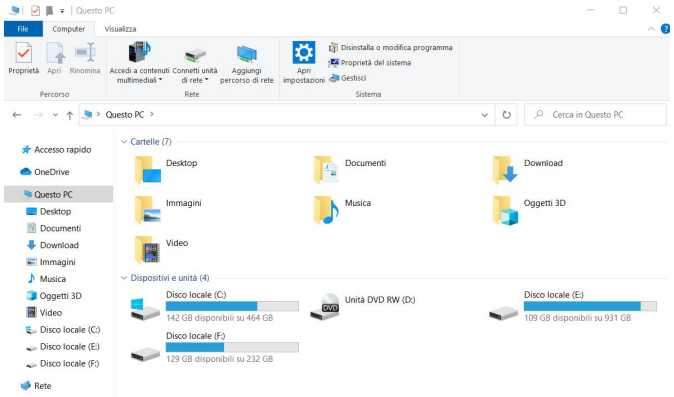
Subjective Organization Schemes

Audience schemes organize content for separate segments of users. Audience schemes can be closed or open, meaning that users are able to navigate from one audience to another. This type of scheme does present challenges unless the content lends itself to users very easily self-identifying to which audience they belong and perhaps not fitting multiple audience profiles.



Metaphor schemes help users by relating content to familiar concepts. This is used in interface design (folders, trash, etc) but can pose challenges when used as the site's primary organization scheme.



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Challenges of Creating Hybrids

Implementing schemes independently has its advantages because it keeps things simple for the user.

They can identify the categorization and form a mental model that can be quickly understood.

Mixing schemes by creating hybrids can cause confusion for users.

This is often proposed as a solution when project teams cannot agree on a single scheme to categorize the content.

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From Principles to Shapes

We've explored *what principles* we use to organize content — alphabetically, by topic, by task, etc.

Now let's look at *what shapes* these organizations take when implemented.

The same organizing principle can take different structural forms:

- Netflix organizes by **topic** (genre) in a **hierarchical** structure
- Wikipedia organizes by **topic** in a **matrix** structure (many interconnected links)

Your choice of structure affects how users navigate and understand the information space.

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Organizational Structures

An organizational structure is how you define the **relationships between** pieces of **content**.

Successful structures allow users to predict where they will find information on the site.

It’s important to take into account user expectations and implement consistent methods of organizing and displaying information so that users can extend their knowledge from familiar pages to unfamiliar ones.

The four main organizational structures are **Hierarchical, Sequential, Matrix and database model.**

source: https://www.usability.gov/how-to-and-tools/methods/organization-structures.html

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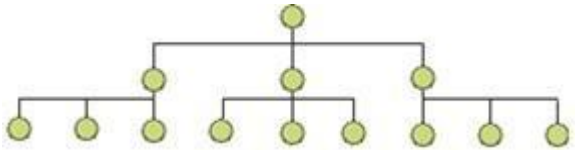
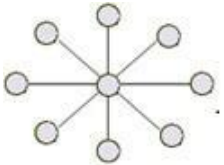
Hierarchical Structures

In Hierarchical Structures, sometimes referred to as tree structures or hub-and-spoke structures, there is a top down approach or parent/child relationships between pieces of information.



Users start with broader categories of information (parent) and then drill further down into the structure to find narrower, more detailed information (child).



Many users are familiar with structuring information in hierarchies because they see these structures on a daily basis in the way businesses have a formed their lead leadership structure, the way project plans are set-up, and so on.

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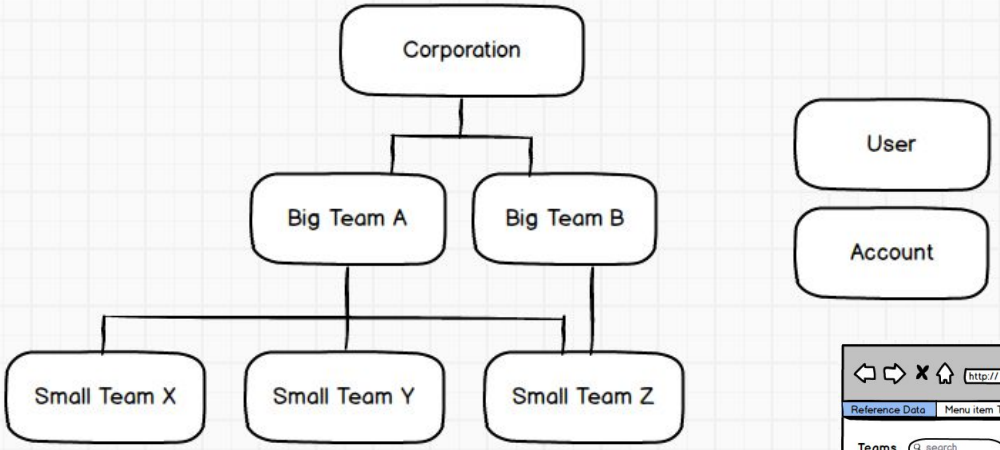

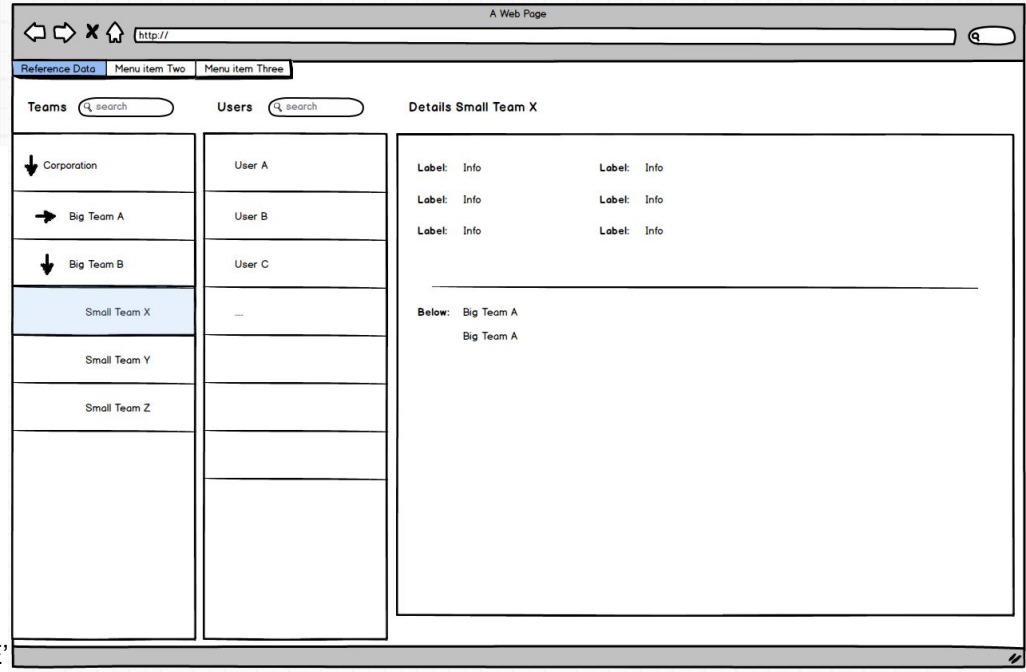

https://ux.stackexchange.com/questions/90872/display-hierarchical-structure-and-corresponding-reference-data

Sequential Structures

Websites with Sequential Structures require users to go step-by-step, following a specific path through content.

An example of this type of structure is when a user is attempting to purchase something or are taking a course online. Sequential structures assume that there is some optimal ordering of content that is associated with greater effectiveness or success.

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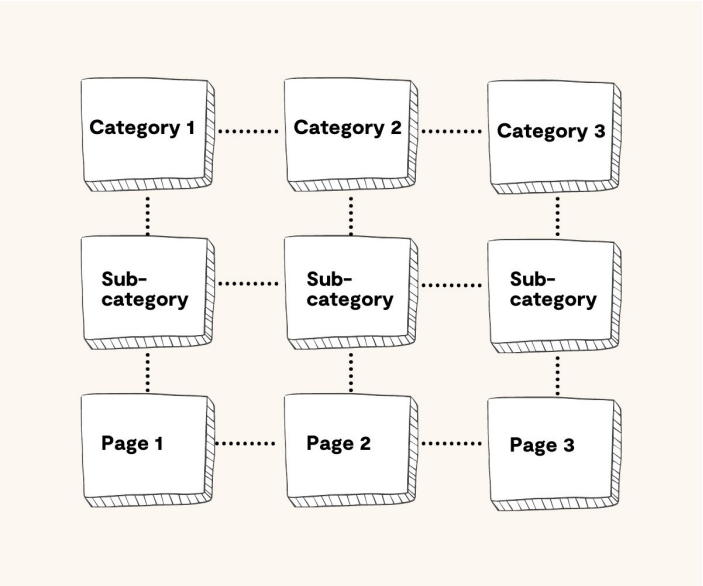
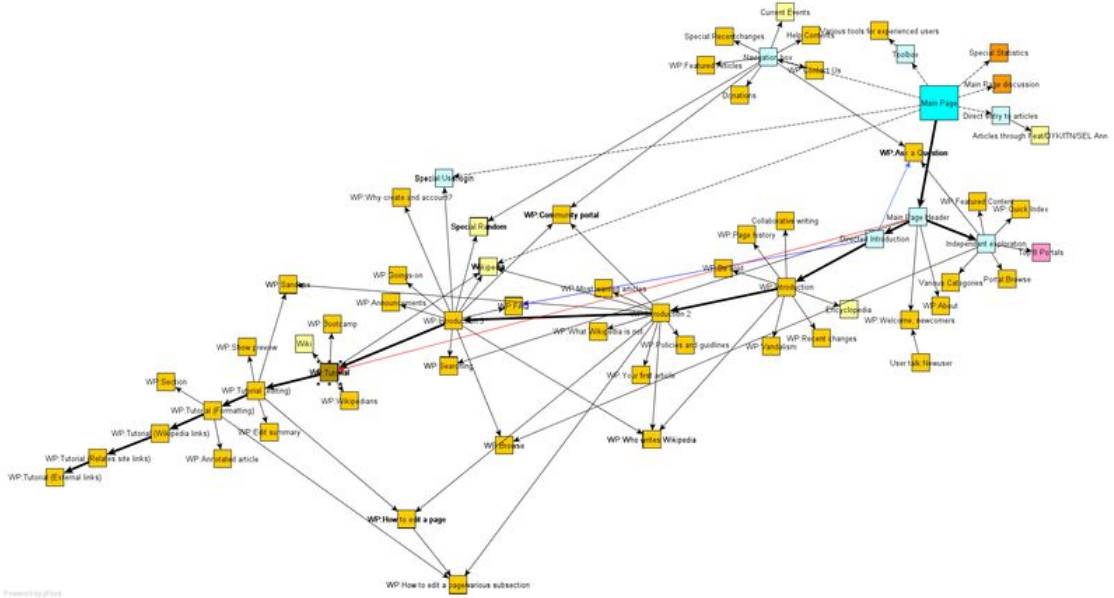
Matrix Structures

A Matrix Structure allows users to determine their own path since content is linked in numerous ways. This type of structure takes full advantage of the principles behind hypertext, or HTML. For example, one user could choose to navigate through a set of content based on date while another navigates based on topic.





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Database Model

The Database Model takes a bottom-up approach. The content within this structure leans heavily on the linkages created through the content's metadata. This type of model facilitates a more dynamic experience generally allowing for advanced filtering and search capabilities as well as providing links to related information in the system that has been properly tagged.

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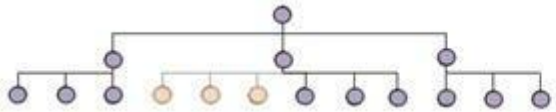
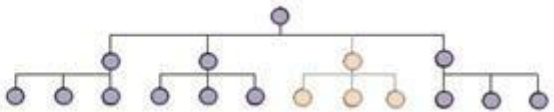
Creating Sustainable Structures

Site/GUI architecture has a long term impact on the site. It's important to put thought into the structure and ensure that it takes into account content updates in the future. Site managers should keep in mind the following when structuring a site:

Allow room for growth. Creating a site that can accommodate the addition of new content within a section (left image) as well as entire new sections (right image).





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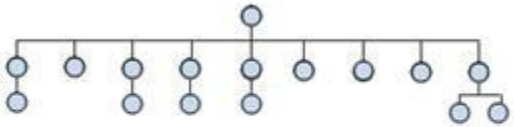
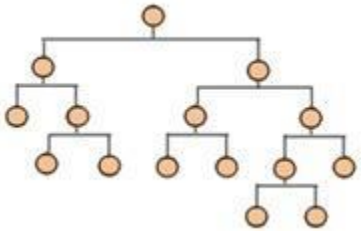
Creating Sustainable Structures

Avoid structures that are too shallow or too deep.

Striking a balance is never easy is an important goal of any architecture. Structures that are too shallow require massive menus.

Users rely on information architects to create logical groupings to facilitate movement throughout the site. In contrast, structures that are too deep bury information beneath too many layers. These structures burden the user to have to navigate through several levels to find the content that they desire.





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Navigation

The study of IA informs the navigation design process.

Navigation is about the **user's orientation in the interface, and its main goal is to enable the user to find the information** and functions he or she is looking for and, not only that, to encourage him or her in a direction that might be desirable to him or her.

The basic principles for good navigation are **findability** and **discoverability**.

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Navigation

Findability means being able to get to the information you are looking for, or more precisely, how the user is able to localize the information he or she considers relevant.

In a library, for example, it indicates the ease with which one can locate a book one is looking for.

Placing elements consistently within the interface and using standards usually promotes findability.

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Navigation

Discoverability refers to the possibility that users have of discovering new information or new features that they were not aware of and were not specifically looking for.

In the library example, it concerns the possibility of discovering a new book that catches one's eye while looking at a shelf or while searching for something else.

The interface often indirectly provides the tools that enable the discovery activity because it can only be evaluated in retrospect. However, it is possible to design for discoverability by placing the new feature, for example, near the most viewed or most relevant items.

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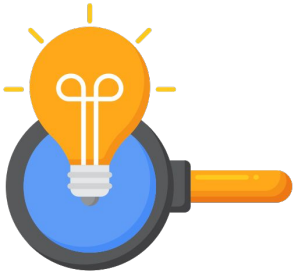
Findability vs Discoverability

With Findability the users knows or expects that a certain content or feature will be available.



With Discoverability the users don’t know that the content or functionality is there: they have to discover it!

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Interfaces' components

When designing your interface, try to be consistent and predictable in your choice of interface elements. Whether they are aware of it or not, users have become familiar with elements acting in a certain way, so choosing to adopt those elements when appropriate will help with task completion, efficiency, and satisfaction.

Interface elements include but are not limited to:

- **Input Controls:** checkboxes, radio buttons, dropdown lists, list boxes, buttons, toggles, text fields, date field
- **Navigational Components:** breadcrumb, slider, search field, pagination, slider, tags, icons
- **Informational Components:** tooltips, icons, progress bar, notifications, message boxes, modal windows
- **Containers:** accordion

Continue: here

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Key Takeaways

- **Design for user behaviors first** Every IA decision should support how users actually think and act (satisficing, spatial memory, safe exploration)
- **Organization schemes define principles, structures define shapes** Choose what categorization logic to use (topic, task, chronological), then decide how to arrange it (hierarchy, matrix, sequential)
- **Good IA is invisible** Users only notice information architecture when it's bad. Your goal is effortless navigation and intuitive organization
- **Balance findability and discoverability** Help users find what they're looking for while enabling them to discover new things they didn't know they needed
- **Consistency enables habituation** Predictable patterns, clear labels, and spatial consistency let users build mental models and work efficiently

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